

Record-019

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Price and then utility function for based on the wait time and maybe energy, right? Yeah. The utility functions. Ah, yes.

So that is right. Yes. So utility function for wait time, I mean the price and the wait time is very straightforward.

You have the mean and the wait time, deviation of the price, mean and deviation of the wait time. Two things you have. Okay.

Yeah. So you can think of, okay, mean is there and you can say create an upper limit and lower limit of the price. Similarly, create an upper limit and lower limit of the wait time.

So mean plus or minus, maybe two sigma or sigma you can use. So the mean plus sigma gives you the upper limit, minus sigma gives you, mean minus sigma gives you lower limit. Now the price preference anyway you have to use a notation for the user.

What is the price preference he has? So there is a mean price. How much is the price per dispatch that the user expects? You choose a random number according to maybe the region where he is present or maybe you can choose any random number of your choice and then you can say given that or maybe you can say based on the trips that were already carried out by the driver you can choose. Then you have a linear function which would be like if the price were to lie in this range that is a lower limit and upper limit.

So you can write it as the price preference minus the lower limit divided by the upper limit minus the lower limit that is the utility function. If the price preference is already above the upper limit you can say the utility is 1. But if it is between the upper limit and the lower limit you can say the price, I mean the utility starts to fall linearly which is the formula I gave you, price preference minus lower limit by upper limit minus lower limit and if the price preference is below the lower limit you can say that utility is 0. In the same way you can write for the wait time and coming to energy what you can do is you can say the utility function in terms of how quickly you are able to complete the charging. So the faster you complete charging the higher the utility and vice versa.

Again a linear utility function can be created and then the last one is the energy based thing. So for that what you can do is whatever charge level that you are choosing you can say that there is a battery degradation cost associated with that you subtract it, so battery degradation cost I think that paper on V to G and G to V something they would have given the function so you can use that paper. So if you see there is a threshold you are having for the, so if I say that you are going to do the charging the faster the charging or the longer the time it takes to charge you can say higher is the utility.

So if you are charging let us say almost you have come down to the threshold, energy threshold, the minimum energy threshold for charging if your energy content is energy is at that level then you can say that your utility is very low but if you are having a sufficient energy which is above the energy threshold then you can say your utility is higher because you will take lesser time to charge. So what I am saying with respect to charging is that you can say that the amount of time you take to charge the smaller it is higher the utility. So you can simply write it as a ratio of the time taken to complete the charging divided by the max time to complete charging that means max time is right from the energy threshold to the full battery.

Similarly, you can say what that depends upon the charger level that you chose. I think we wrote it last time know that is the what is the desired charging you want energy you want in the battery minus what is the lower threshold or where what is the current SOC divided by the number of divided by the charging value suppose if it is chosen as let us take an example suppose if the energy threshold is 20% of the SOC and the current I mean 20% or your SOC is 20% suppose and your battery capacity full battery capacity you want to do then you are looking at it like 0.8 is the amount of SOC you have to charge. So 0.8 times the battery capacity if you write you will get how much energy you have to now charge or store correct.

Suppose if you say you are at 40% capacity right now your SOC is at 40% then it is like you have to store about 60% energy. So we will assume that if you are going for charging you will complete the charging fully. So 60% of the battery capacity you have to store.

Now what is the charger level you are choosing that charger level will be maybe you can say for example if it is 6 kilowatt hour you have chosen then the number of slots that you will take to charge will be suppose you are at 40% SOC when you are starting to charge then it is like what 60% SOC you have to charge so 60% into battery capacity divided by 6 kilowatt hour so that will give you the number of slots number of hours it takes to charge. You divide the number of hours by the slot duration you will get the number of slots. Am I clear? Yes sir.

Okay. If you are going to do the slowest charging you will incur the largest number of slots. If you are going to do fast charging you are going to incur the lesser number of slots.

So for example if you say that I have 1 minus of the number of slots to charge divided by the max number of slots that can be a utility function. So the faster you charge the higher is the utility right then you also have a battery degradation cost. The battery degradation cost acts the opposite way meaning if you choose a higher charge level charging level then the battery degradation cost is also higher and if your charger level is chosen as the lowest level your battery degradation cost will be lowest.

So take the difference of these two and then call that as the utility of charging. Utility of charging okay. So it is like a weighted sum of the utility of having a certain amount of energy or utility of storing a certain amount of energy minus the degradation cost.

What could be the weights here sir? You just take equal. Okay. No need to write any weighting factors for any of these.

As I told you if you introduce weighting factors and they will ask you to do experiments for different weighting factors. So they will say do a sensitivity analysis it will take a lot of your paper space. So we will assume that all the weights are equal.

So here we are calculating utility function for pricing, for weighting and for charging. Yes. Okay.

Yes sir. Got it. Pricing, weighting and charging.

Yes. Understood sir. Okay.

G plus weighted. Okay sir. That is all.

So I think today I will be completing the tabulation, I will be completing it by afternoon and then I will, I have to make the tabulation a little more better because right now I have written some description and all that I have to replace it with only tick and cross. Once I am done with this I think today I will start the introduction section. I will try to write it.

So from my collection of papers here what I see is there are if I say I want 20-20 papers.

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